DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM
January 2014
ALGEBRA (Ph.D. Version)

Instructions to the student

a. Answer all six questions; each will be assigned a grade from 0 to 10.

b. Use a different booklet for each question. Write the problem number and your **code number (not your name)** on the outside of the booklet.

c. Keep scratch work on separate pages in the same booklet.

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1. **(a)** Let G be a finite group and let K be a normal subgroup. Assume that $i = [G : K]$ and $|K|$ are relatively prime. Show that

$$K = \{x^i | x \in G\}.$$

(You must indicate where you use the assumptions of normality and relative primality.)
(b) Give an example to show that the conclusion of part (a) can be false if K is not assumed to be normal in G.

2. Let T be a linear transformation of a finite-dimensional complex vector space V such that $T^2 = T$. Let $\alpha \neq 1$ be a complex number. Show that $I - \alpha T$ is invertible, where I is the identity map of V.

3. **(a)** Show that the ideal $I = (3, X^6 + 1)$ is not a prime ideal of $\mathbb{Z}[X]$.
(b) Find prime ideals $A \neq 0$ and B such that $A \subset I \subset B \subset \mathbb{Z}[X]$. Prove that the ideals you find are prime ideals.

4. Let R be a commutative ring with 1. An R-module F is called *flat* if, whenever $f : M \to N$ is an injective homomorphism of R-modules, the induced map $M \otimes_R F \to N \otimes_R F$ is also injective. Suppose R is a principal ideal domain. Show that a finitely generated R-module is flat if and only if it is torsion free.

5. Let $K/\mathbb{Q}$ be a Galois extension of fields ($\mathbb{Q}$ denotes the rational numbers) with $[K : \mathbb{Q}] = 21$. Assume that $G = \mathrm{Gal}(K/\mathbb{Q})$ is non-abelian. Let $f(X) \in \mathbb{Q}[X]$ be an irreducible polynomial of degree 7 whose roots $\alpha_1, \alpha_2, \ldots, \alpha_7$ lie in K.
(a) Show that the splitting field of $f(X)$ is K.
(b) Show that there are exactly 7 distinct fields F with $\mathbb{Q} \subset F \subset K$ and $[F : \mathbb{Q}] = 7$, and for each pair F_1, F_2 of such fields, there is a $g \in G$ such that $g(F_1) = F_2$.
(c) Show that each of the fields F in part (b) has the form $F = \mathbb{Q}(\alpha_i)$ for some $1 \leq i \leq 7$.

6. Let G be a finite group and let $R = \mathbb{C}[G]$ be the group ring of G with complex coefficients. We may regard R as a complex vector space of dimension equal to the order of G. Consider the representations $\rho_i : G \to GL(R)$, $i = 1, 2, 3$, defined by

$\rho_1(g)(r) = grg^{-1}$,
$\rho_2(g)(r) = gr$,
$\rho_3(g)(r) = rg^{-1}$,

where $g \in G$ and $r \in R$.
(a) Show that the representations ρ_2 and ρ_3 are equivalent.
(b) Show that ρ_1 is equivalent to ρ_2 if and only if G is trivial.
(c) Suppose G is abelian and let $\rho_4(g)(r) = g^2 r$. Show that ρ_2 is equivalent to ρ_4 if and only if G has odd order.